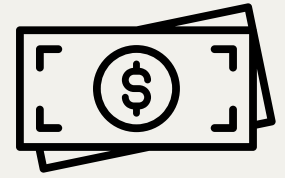


Prediction of the Shadow Economy in Eastern Europe



Author:
Federica Keni

Advanced Data Analysis 2025
University of Lausanne

A Machine Learning Approach with Regional Heterogeneity Analysis

What is Shadow Economy?

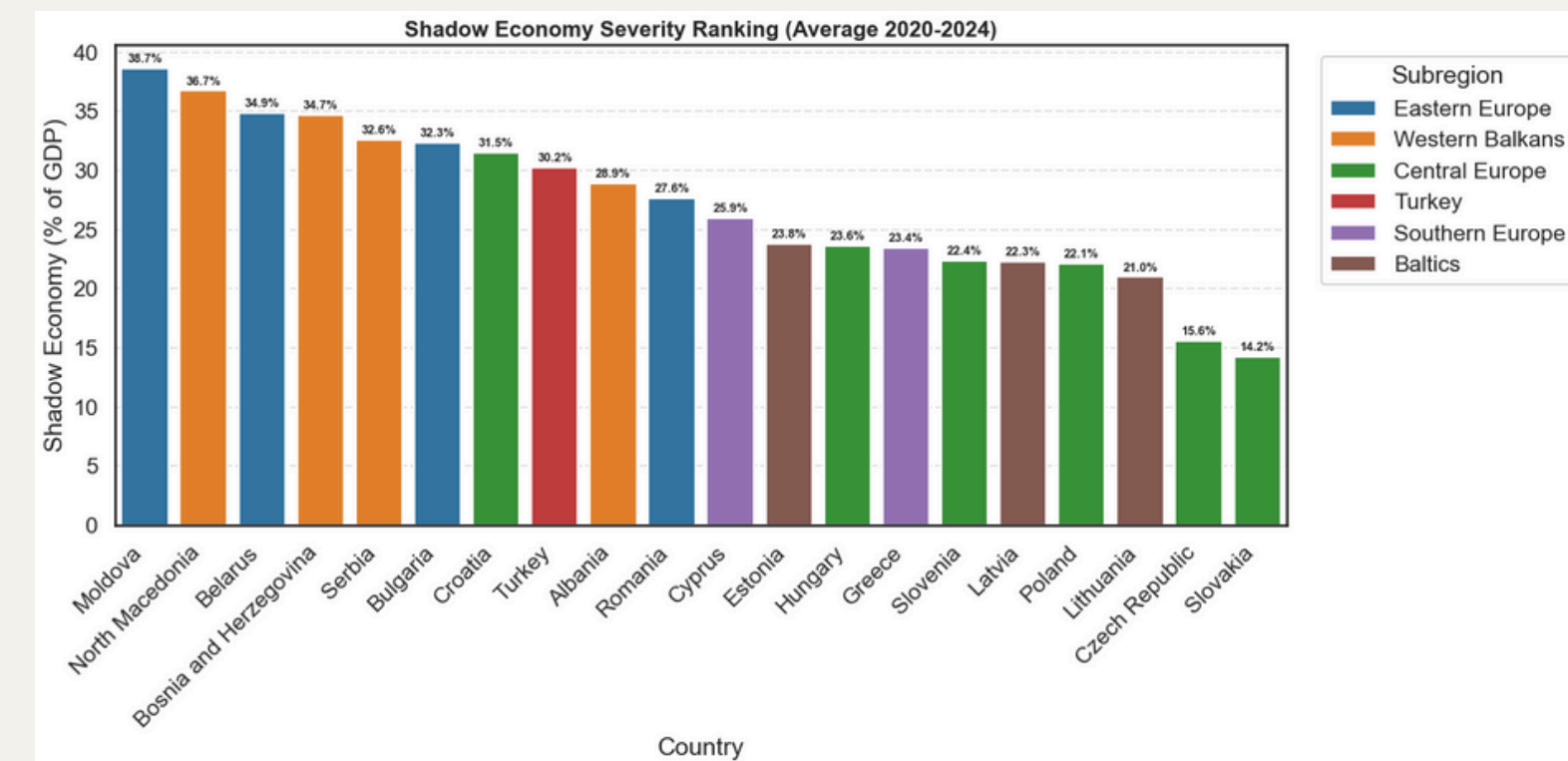
- “Legal economic activities hidden from authorities to avoid taxes or regulations”.
- Massive structural barrier to economic convergence in Eastern Europe.

The Problem:

- Existing econometric models rely on history (**INERTIA**)

The Objective:

- The Shift: Moving from **estimating past levels** to guiding future policy.
- Methodology: A **Supervised Machine Learning** regression to isolate structural drivers from persistence.



Author's calculations based on World Bank & IMF data (2020-2024).

The Literature Gap

Understanding the limitations of current models

Current Standard: MIMIC Models

Multiple Indicators Multiple Causes (IMF Working Paper, 2019)

$$ShadowEconomy_{i,t} = \alpha_i + \beta X_{it} + \delta_t Time_t + u_{i,t}$$

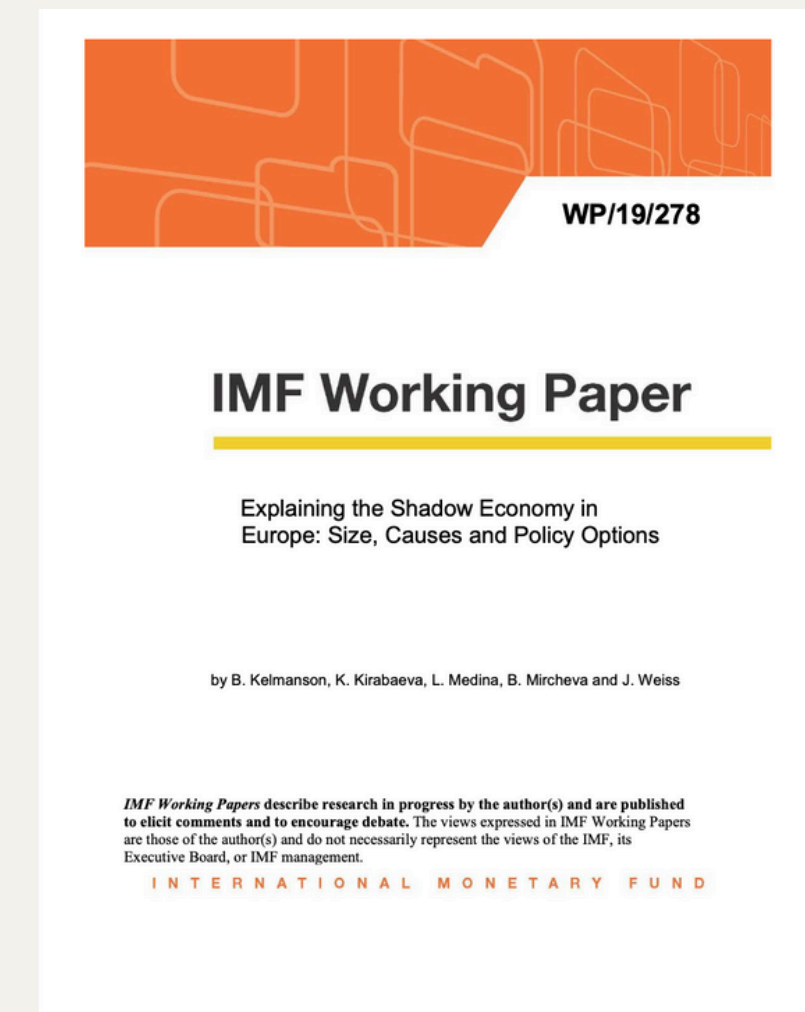
where *Shadow Economy*_{*i,t*} represents the size of the shadow economy as a share of GDP; α_i are country fixed effects; $X_{i,t}$ is a vector of macroeconomic variables and institutional indicators; $Time_t$ is time fixed effects, which we include to control for unexpected year-related variation and special events; and $u_{i,t}$ is the error term. We start with a group of

✓ Strengths

- Robust historical estimates (ex-post)
- Theoretically grounded
- Long-term time series

✗ Critical Weakness

- **Highly inertial & smooth→ latent variable**
- Best predictor: today's level
- Cannot capture regime shifts
- Limited ex-ante foresight



Research Questions

Key inquiries guiding the study's approach



RQ1 (Inertia Test)

Can **ML** algorithms utilizing **expanded** institutional, digital and economic features outperform a naive persistence baseline in predicting shadow economy levels?

RQ2 (Fundamental Test)

Can **fundamentals alone** (excluding lagged shadow economy values) predict levels competitively, or is the inertial component dominant?

RQ3 (The Structural Horizon Test)

Does shifting the target to a 3-Year Change allow the model to **capture structural transitions** better than annual predictions?

Data Overview & Variables

Dataset Overview:

- 20 **Eastern European Countries** and adjacent countries | 2000-2024
- N =500 observations (20 countries * 25 years)
- **Subregions:** Western Balkans: Eastern Europe; Central Europe; Southern Europe; Turkey

Target Variable: Shadow Economy (% of GDP)

Source: MIMIC estimates (IMF methodology, 2019)

Explanatory Variables:

Institutional Quality

- Governance_Index

Economic variables

- GDP_per_Worker
- Tax Revenue_GDP%
- Agriculture_GDP%

Human Capital & Technology

- Human_Capital_Index
- Digitalization_Growth_3yr

Macroeconomic variables

- Trade_Openness_GDP%
- Remittances_GDP
- Inflation Rate (%)
- Unemployment Rate (%)

Feature Engineering

Governance Index (PCA):

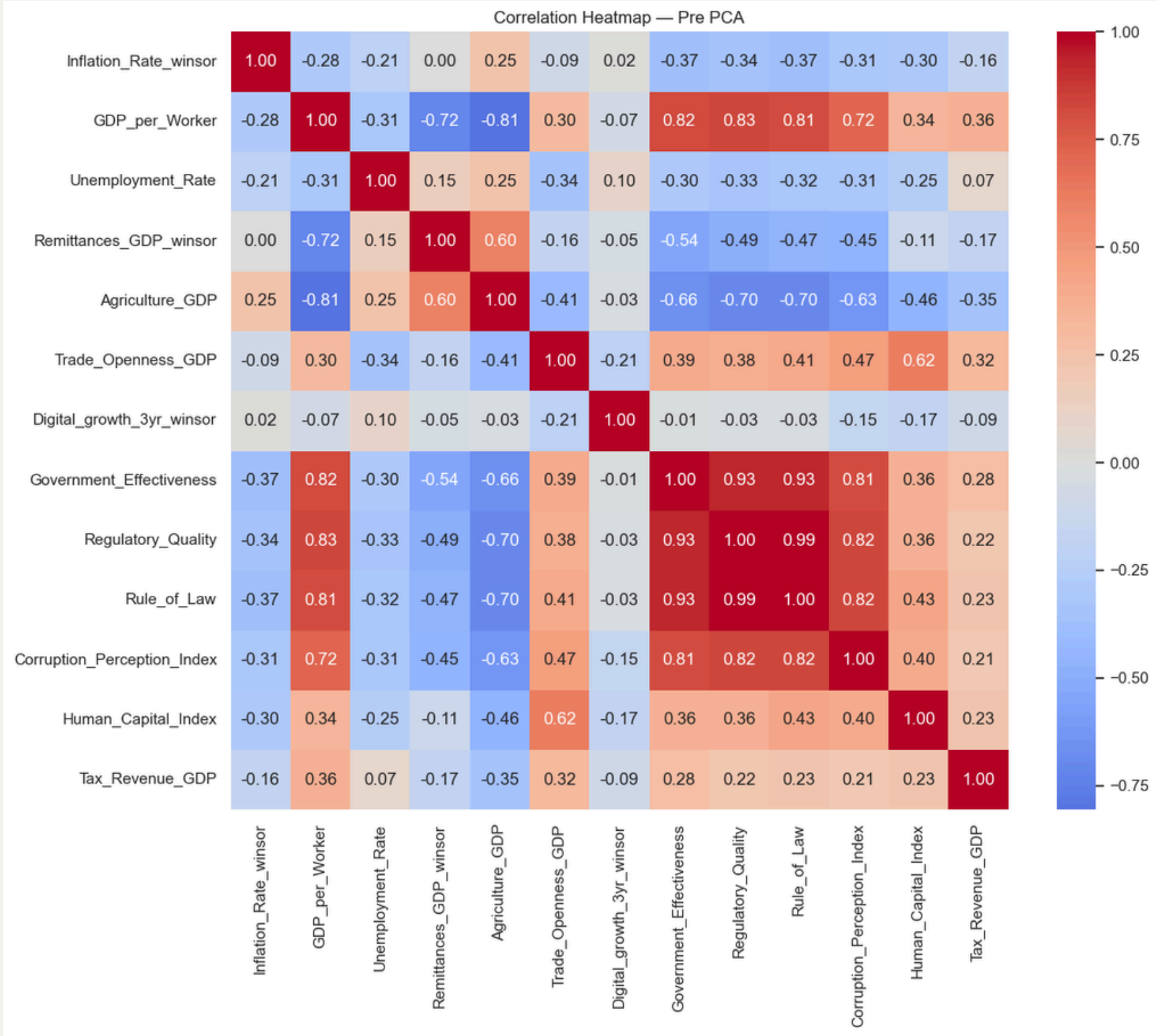
- Government Effectiveness,
- Rule of Law,
- Regulatory Quality;
- **Corruption Perception Index-> NEW**

Tax Revenue/GDP (%)

- Role: Proxy for Fiscal Pressure
- Handling Missing Data: **PMM**

Digitalization Growth (3 Year)

- Trend in Internet Adoption
- Proxy for transaction traceability



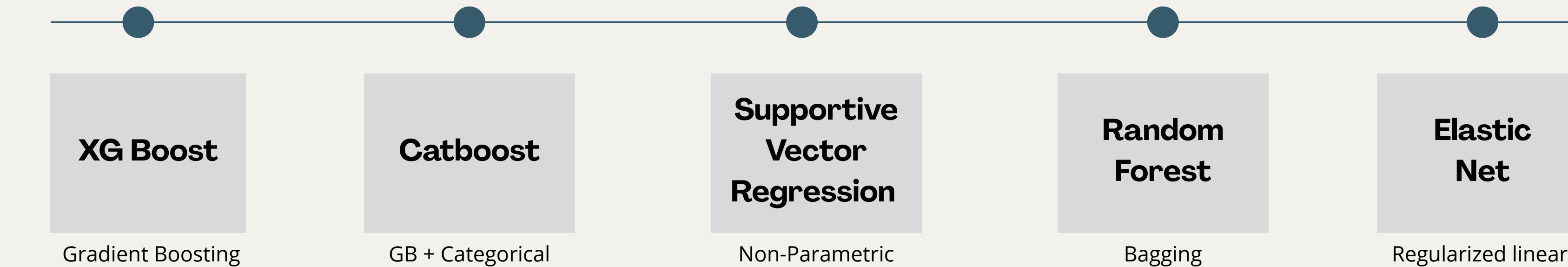
Validation Methodology

Temporal Validation Framework:

- Initial training (2000-2015)
- **Expanding-Window Walk-Forward (2016-2023)** —————>
- Hold-out Test (2024) → retrained on $t < 2024$

TRAIN/TEST SPLIT BY VALIDATION YEAR			
Test_Year	Training_Years	Training_Observations	Test_Observations
2016	2000-2015	320	20
2017	2000-2016	340	20
2018	2000-2017	360	20
2019	2000-2018	380	20
2020	2000-2019	400	20
2021	2000-2020	420	20
2022	2000-2021	440	20
2023	2000-2022	460	20

Five Algorithms Benchmarked:



The Three-Model Architecture

Model 1

(Level with Inertia):

Predict **annual level** using
lagged fundamentals: $X(t-1)$ +
lagged shadow economy $Shadow_{t-1}$.

- **Naive Baseline (Inertia):**

$Shadow_t = Shadow_{t-1}$

Model 2

(Fundamentals Only):

Predict **annual level** using only
fundamentals **$X(t-1)$** , without
considering the lagged shadow
economy

- **Naive Baseline (Inertia):**

$Shadow_t = Shadow_{t-1}$

Model 3

(3-Year Change):

Predict **$\Delta Shadow_{3yr} = Shadow_t - Shadow_{t-3}$** using fundamentals
only.

- **Zero Change Model Baseline:**

$\Delta Shadow_{3yr} = 0$ in year t

Evaluation Metrics:

- **RMSE:** Root Mean Squared Error
- **MAE:** Error Magnitude
- **Improvement %:** Performance vs baseline
- **Sign Accuracy:** Did we predict direction correctly?



Results: The Barrier of Inertia (Models 1 & 2)

Model 1 & 2 Performance: Negative Findings

Conclusion:

- No algorithm beats **persistence** for annual levels.
- **Fundamentals** alone **cannot** predict size without history.

MODEL 1: PERFORMANCE COMPARISON TABLE						
Algorithm	RMSE_Test	MAE_Test	R2_Test	Improvement_%	Sign_Accuracy_%	
Naive Baseline	1.925460	1.418841	NaN	0.000000	NaN	
Elastic Net	1.931500	1.490918	0.884616	-6.411921	55.000	
CatBoost	2.168706	1.729124	0.893103	-49.658589	56.250	
SVR	2.214793	1.719860	0.872575	-38.490045	56.250	
Random Forest	2.308133	1.771851	0.869585	-50.211707	53.125	
XGBoost	2.497354	1.983797	0.836933	-66.959731	53.125	

MODEL 2: PERFORMANCE COMPARISON TABLE						
Algorithm	RMSE_Test	MAE_Test	R2_Test	Improvement_%	Sign_Accuracy_%	
Zero-Change Baseline	1.925460	1.418841	NaN	0.000000	NaN	
CatBoost	1.980798	1.562252	-0.789815	-3.380812	56.250	
Elastic Net	1.988164	1.580114	-0.799898	-3.969922	55.000	
Random Forest	2.159575	1.621442	-1.583973	-19.376193	53.750	
SVR	3.221544	2.513000	0.769032	-127.367818	57.500	
XGBoost	4.389655	3.520501	0.610634	-261.106459	54.375	

Short-term levels are sticky

To see the structural trends -----> Shift the horizon (Model 3)



Results: Success of Model 3 (3-Year Change)

Catboost dominates:

- 24% improvement
- Lower RSME(2.50 vs 3.00) and MAE (1.96 vs 2.12) than baseline
- High Regional Heterogeneity

MODEL 3 -Validation (2016-2023)

MODEL 3: PERFORMANCE COMPARISON TABLE					
Algorithm	RMSE_Test	MAE_Test	R2_Test	Improvement_%	Sign_Accuracy_%
SVR	1.250673	0.951457	-2.700368	-37.032147	61.0
CatBoost	2.505281	1.961968	-0.073930	23.950939	65.0
Random Forest	2.927250	2.262480	-0.618418	8.458310	63.0
Zero-Change Baseline	3.002031	2.127240	NaN	0.000000	NaN
Elastic Net	3.407935	2.657411	-1.319895	-13.120941	57.0
XGBoost	3.532667	2.753982	-2.015065	-22.830652	60.0

Catboost Model 3 -Validation (2016-2023)

SUB-REGIONAL PERFORMANCE			
Sub_Region	RMSE	MAE	N_Observations
Baltics	4.420	2.638	15
Central Europe	2.040	1.585	30
Turkey	4.627	2.271	25
Western Balkans	2.247	1.743	30

Robustness (2024 Hold-Out):

- RSME(1.801 vs 1.805)
- MAE (1.328 vs 1.473)
- 75% Sign Accuracy (Direction) (R² 0.919)



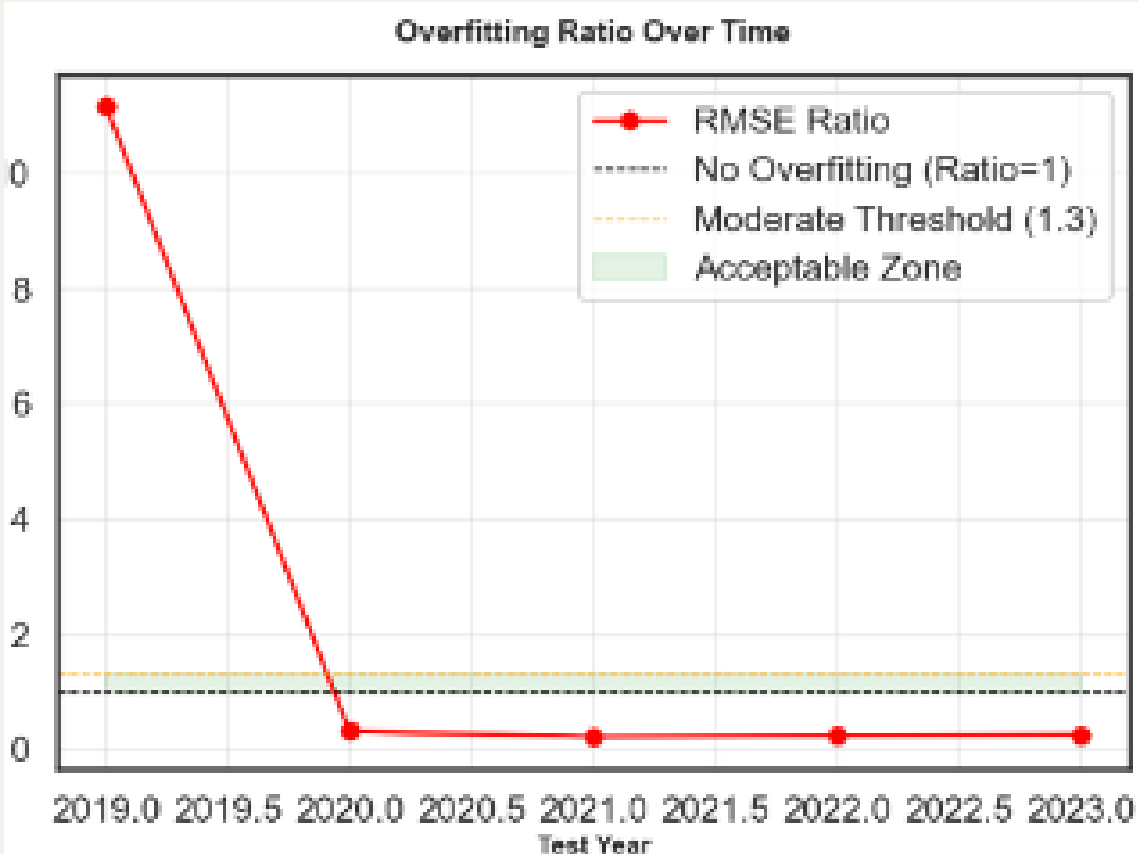
Hyperparameter Tuning

Why Tuning Matters Challenge: Small panel (500 observations)-> Overfitting Risk
Solution: GridSearchCV with TimeSeriesSplit

Optimal Parameters (Catboost)	Value
Tree Depth	6
Learning Rate (<i>lr</i>)	0.05
L2 Leaf Regularisation	5
Iterations	100
Subsample	0.8

Noise Filter:

- Low *learning rate* (0.05) ignored annual noise
- Adapting to 2019 Breakthrough



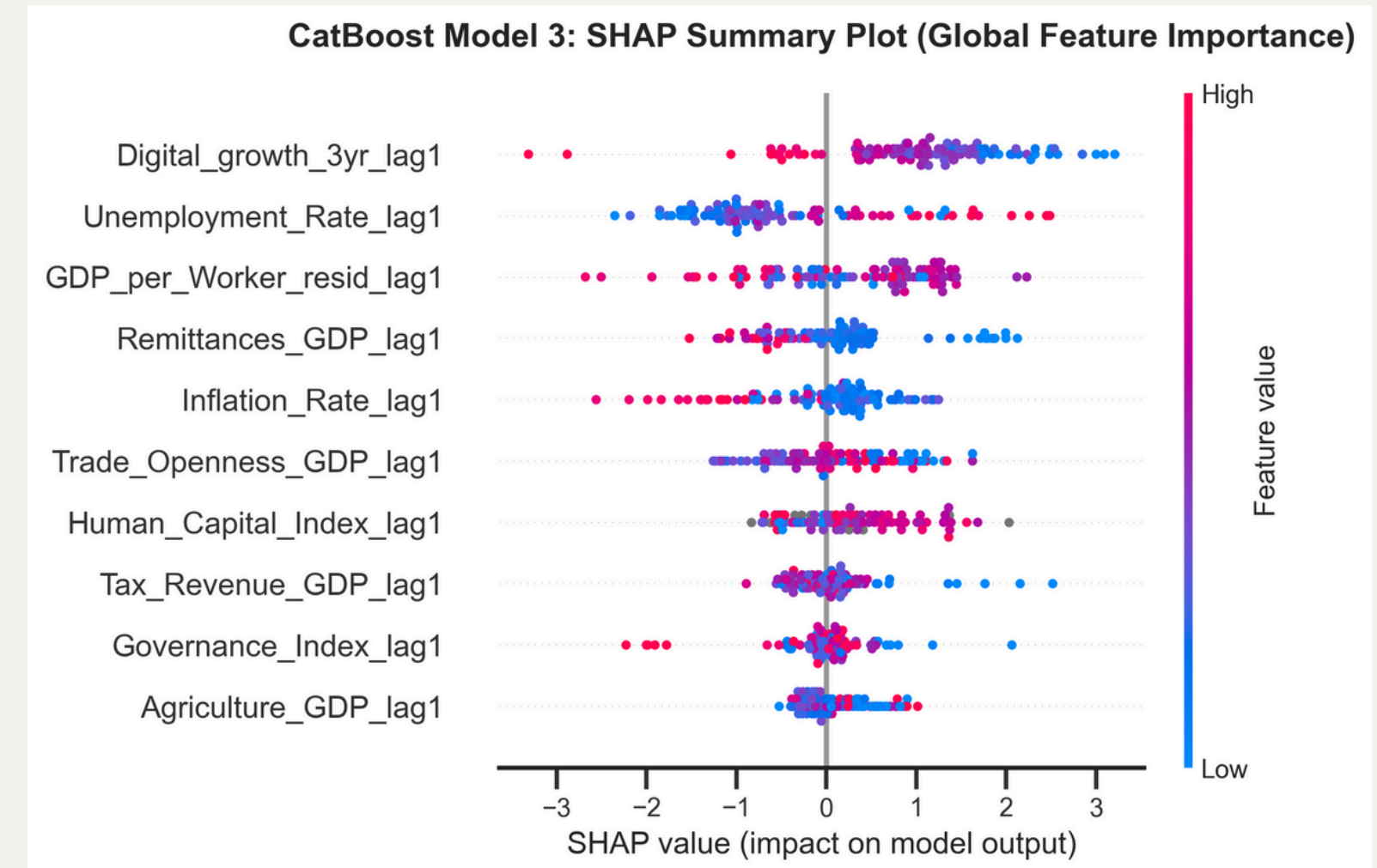
Feature Importance Analysis

Digital Dominance:

- Digital_Growth_3yr as #1 predictor of structural change
- Mean (SHAP 1.27) is 4x more predictive than traditional Tax Policy (SHAP 0.31)

Top 10 Features by Mean |SHAP Value|:

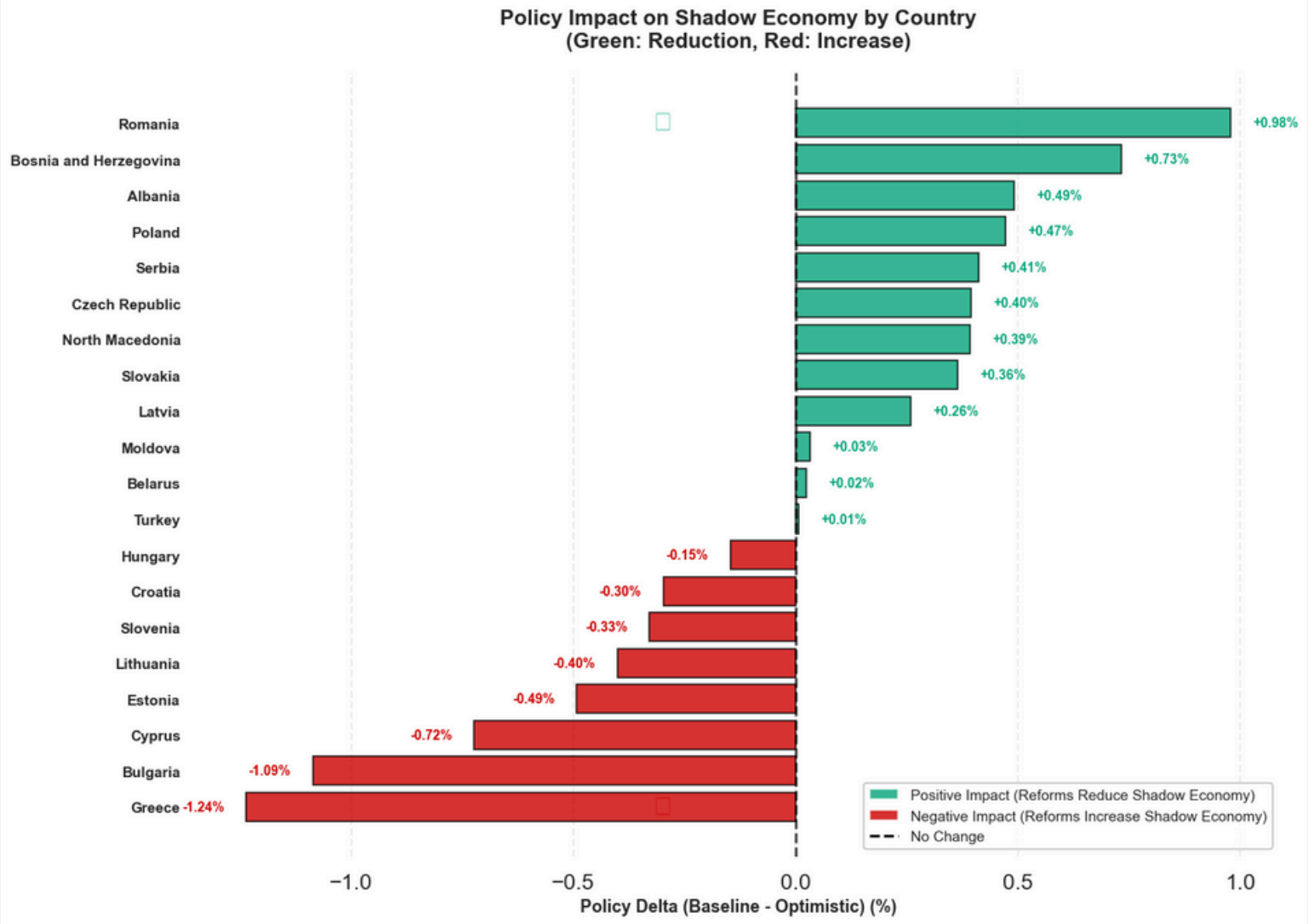
Feature	Mean_ SHAP_Value
Digital_growth_3yr_lag1	1.271452
Unemployment_Rate_lag1	1.022806
GDP_per_Worker_resid_lag1	0.835296
Remittances_GDP_lag1	0.583573
Inflation_Rate_lag1	0.563319
Trade_Openness_GDP_lag1	0.557292
Human_Capital_Index_lag1	0.541731
Tax_Revenue_GDP_lag1	0.311542
Governance_Index_lag1	0.311448
Agriculture_GDP_lag1	0.310337



Regional Heterogeneity & Scenario Analysis

Region	Top Driver	Importance
Baltics	Digital Growth (3 Year)	41.2%
Western Balkans	Remittances	32.9%
Central Europe	Governance	22.6%
Turkey	Human Capital	29.0%
Southern Europe	Unemployment Rate	19.4%

Policy Impact in Eastern Europe



X Structural Inertia dominates: Scenario analysis proves that isolated, short-term shocks generate negligible change (-0.008%), **failing to overcome 25 years of structural inertia.**

Methodological Conclusions

Validation of
Inertial Dominance
(The "Negative" Result)

Success of the
"Horizon-Shift"
Architecture

Non-Linear Feature
Discovery (SHAP)

- Benchmarked ML vs. L vs. Naive Persistence (t+1): All models failed to beat the baseline ($R^2 \leq 0$).
Conclusion: Short-term data is **dominated by inertia**, not fundamentals.
 - Shifted target to 3-Year Structural Change: CatBoost achieved +24% RMSE Improvement and 75% Sign Accuracy.
Conclusion: Structural signals only emerge at multi-year horizons.
 - Digital Growth (25%) > Tax Policy (10%)
The Win: Captured **non-linear threshold effects** that linear econometrics (MIMIC) miss
- > **Model Constraints:** Small-N Sample (~500 obs) required Ordered Boosting to prevent overfitting;
sensitive to 2019 structural break.

